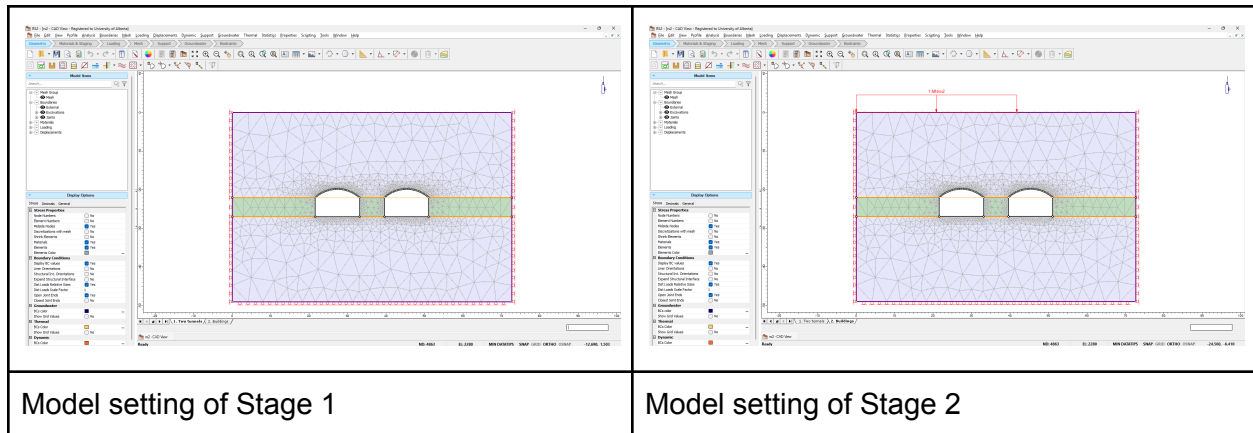
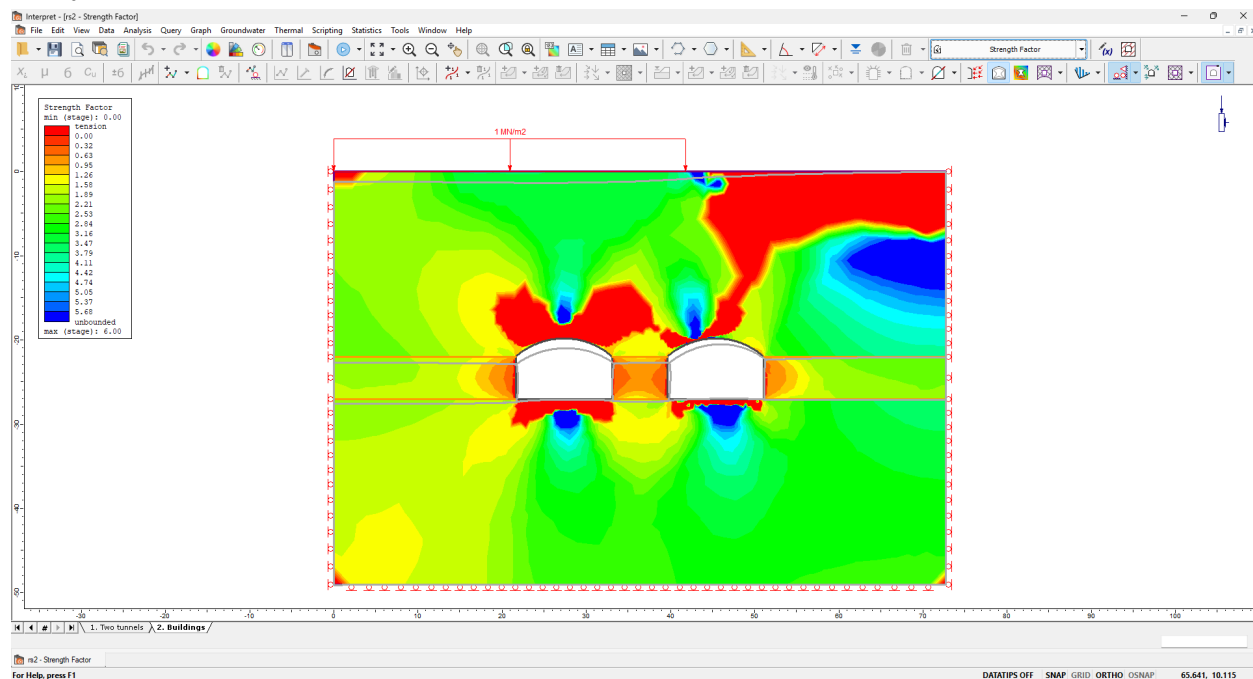


Question 1 of Homework 4

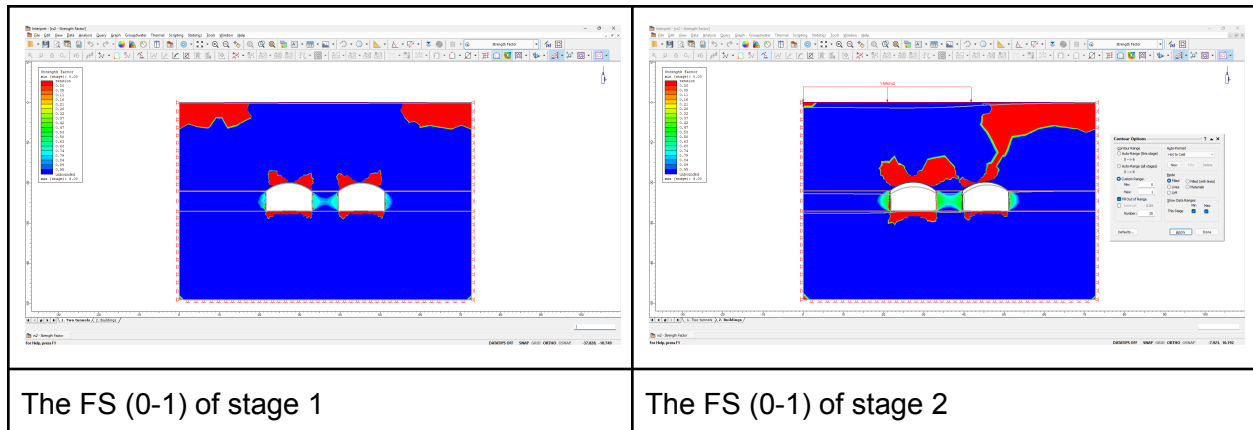
The model setting is shown as follows:



After computing, the analysis results can be obtained from the interpretation. The FS (factor of safety) is shown below:

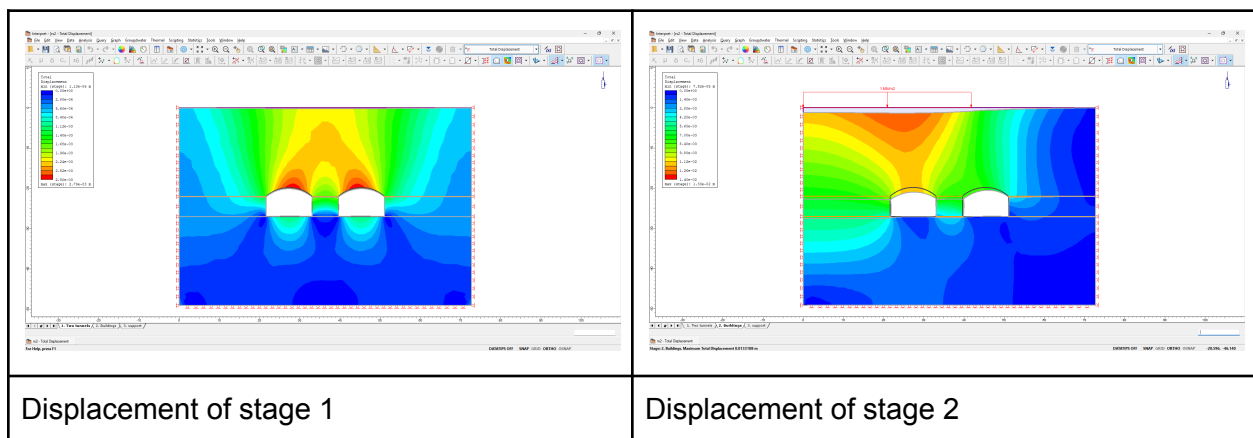


By setting the contour range to 0-1, the dangerous areas can be easily observed, as shown in the figures below.



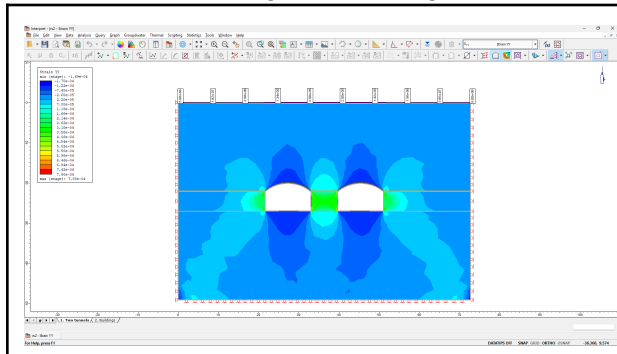
Based on this, these two tunnels are unstable. There are massive red areas ($FS < 1$) existing in stage 1 and stage 2, which means these two tunnels will fail.

Changing the analysis to displacement, we found that the rock mass underneath the building and the tunnels will generate relatively large vertical displacement, as shown below.

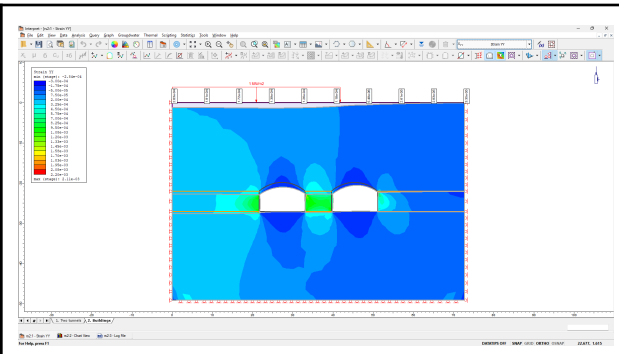


Question 2

The strain YY of stage 1 and stage 2 is shown below

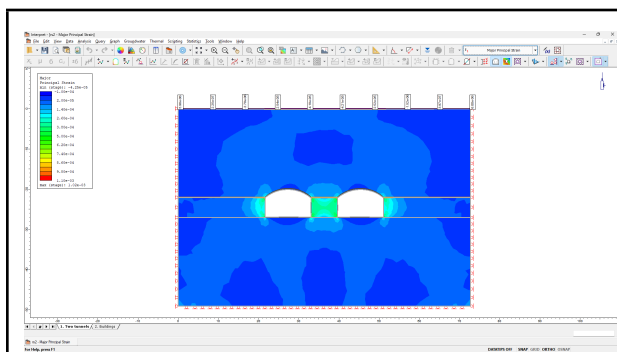


The strain YY of stage 1

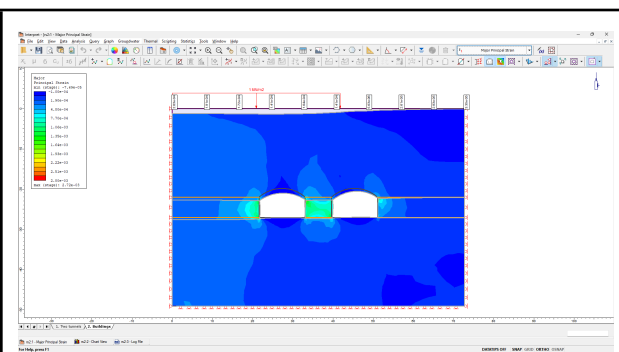


The strain YY of stage 2

As we can see from the strain YY, the values of the strains (YY) underneath the building are significantly larger than those values in stage 1. The maximum strain YY underneath the building is 2.03×10^{-4} , which is located in the left upper corner.

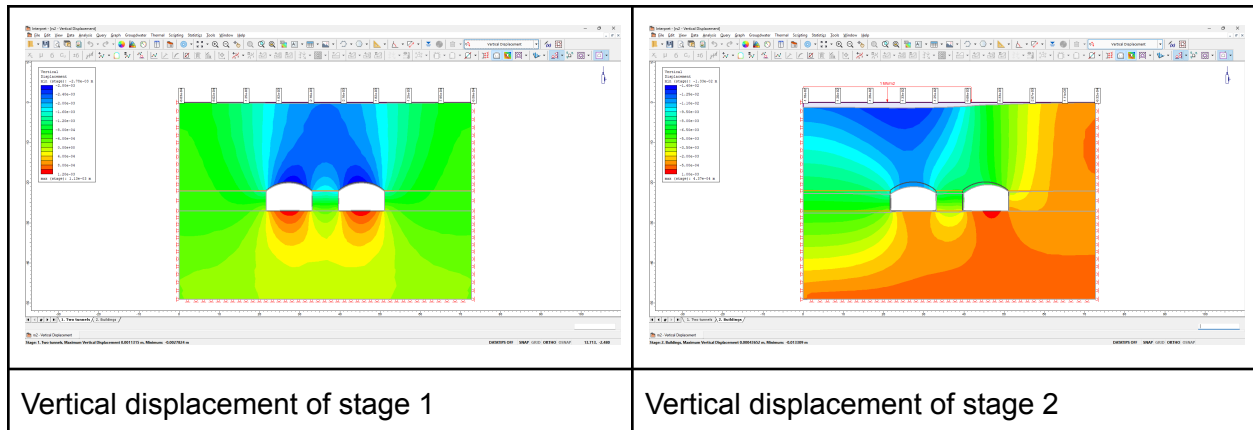


Major principal strain of stage 1

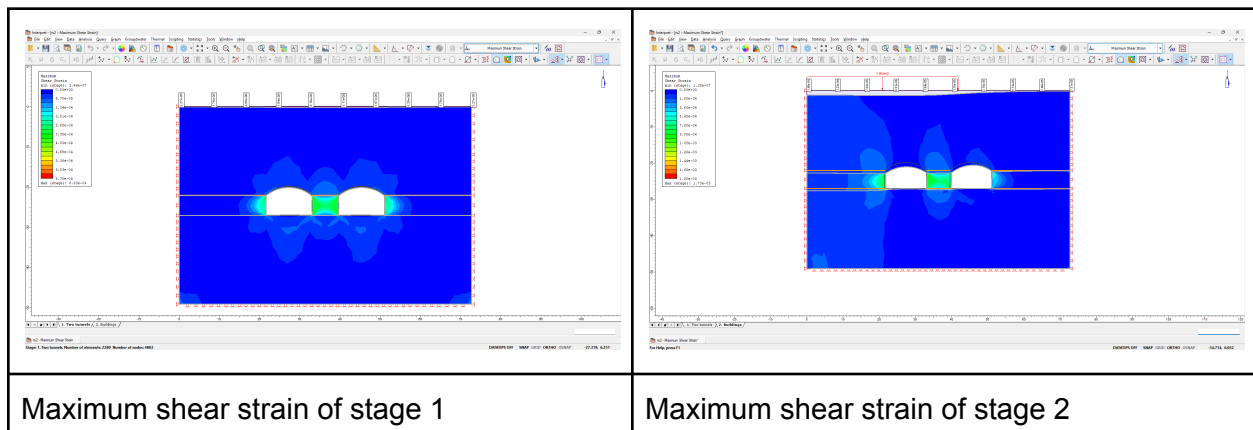


Major principal strain of stage 2

Similarly, as we can see from the major principal strain, the values of major principal strain underneath the building are significantly larger than those values in stage 1. The principal strain under the building is 2.03×10^{-4} , located in the left upper corner and equal to the maximum strain YY.



Similarly, for vertical displacement, as we can see from above, the values of vertical displacement underneath the building are significantly larger than those values in stage 1. The vertical displacement underneath the building is $-1.18\text{e-}2$, which is located in the left upper corner and equals the maximum strain YY as well.



Same results for maximum shear strain, the maximum value of maximum shear strain underneath the building is located at the left upper corner, $1.43\text{e-}4$.

Question 3

These two tunnels are totally unstable, and some necessary support measures should be implemented, such as adding bolts or lining the tunnel surface.

The default liner 1 properties:

Liner Properties

Liner: Liner 1

Color	
Liner Type	Standard Beam
Formulation	Timoshenko
Thickness	0.1 m
Elastic Properties	
Young's modulus	30000 MPa
Poisson's ratio	0.2
Axial Strain (+Expansion)	0

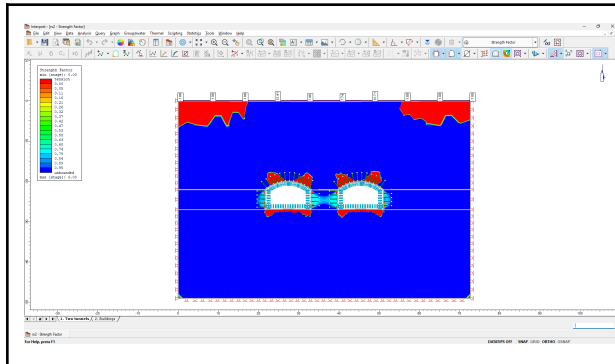
The default bolt 2 properties:

Bolt Properties

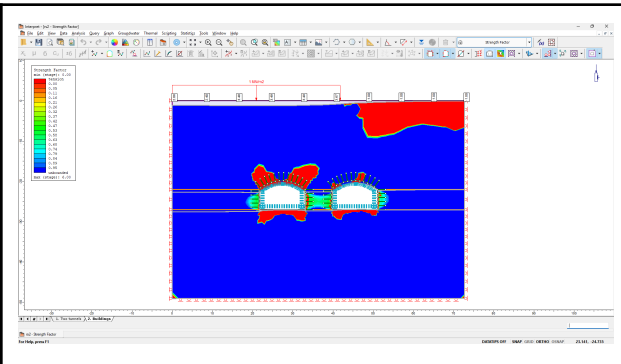
Bolt 2

Bolt Color	
Bolt Type	End Anchored
Bolt Diameter	19 mm
Bolt Modulus,E	200000 MPa
Tensile Capacity	0.1 MN
Residual Tensile Capacity	0 MN
Out-of-Plane Spacing	1 m
Pre-Tensioning Force	0 MN
Constant Pre-tensioning Force in Install Stage	Yes

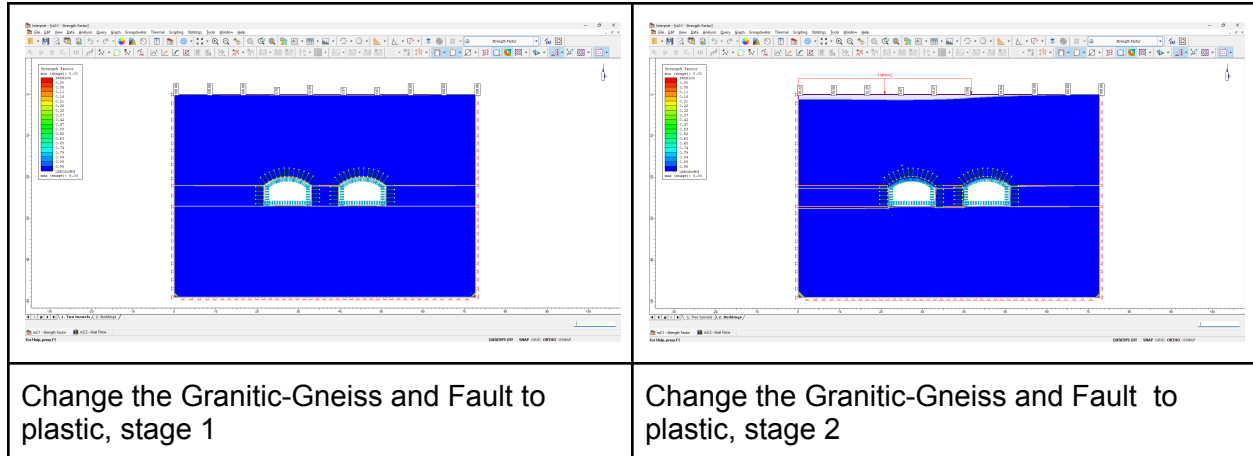
However, though the bolts and liners are both applied, the results show that the left upper and right upper corners are still unstable. They are only a little bit affected by the tunnels (Compare with the non-adding bolts and liners).



The FS of stage 1 (after adding supports)



The FS of stage 2 (after adding supports)



The above figures I got after changing the material to plastic, and they showed that everywhere is safe. Actually, it still doesn't make sense, because I noticed that the residual cohesion is larger than peak cohesion in the default setting. Even though I tried to change it to a more reasonable number, the RS2 reports that some error occurred, and it can not give the correct answer.

Therefore, I guess that due to our rock mass and the fault being too weak, we assume they are elastic, and they will not generate plastic deformation. So, the liners and bolts can not play a role in this case.